

Eenheid 1.3 Wat is 'n funksie?

Note Title

2015/01/26

(Studiegids p9)

Het jy geweet?

- Leibniz (1646 – 1716) was die eerste wiskundige wat die term **funksie** gebruik het om te verwys na afhanklike hoeveelhede en uitdrukkings.
- Euler (1707 – 1783) was verantwoordelik vir die moderne notasie **$f(x)$** om **funksie waardes** aan te dui

Daar is 4 maniere om in funksie
voor te stel:

- woorde
- visueel (grafieke)
- tabel (numeriese)
waarde
- algebraïes (formule)

Funksies :

- Bepaal gebied
- Skets funksie
- Bepaal dan waarde-
versameling

Gebied =
Definisieversameling

Onthou: $f : D_f \rightarrow W_f$

① $f(x) = \sqrt{2-x}$

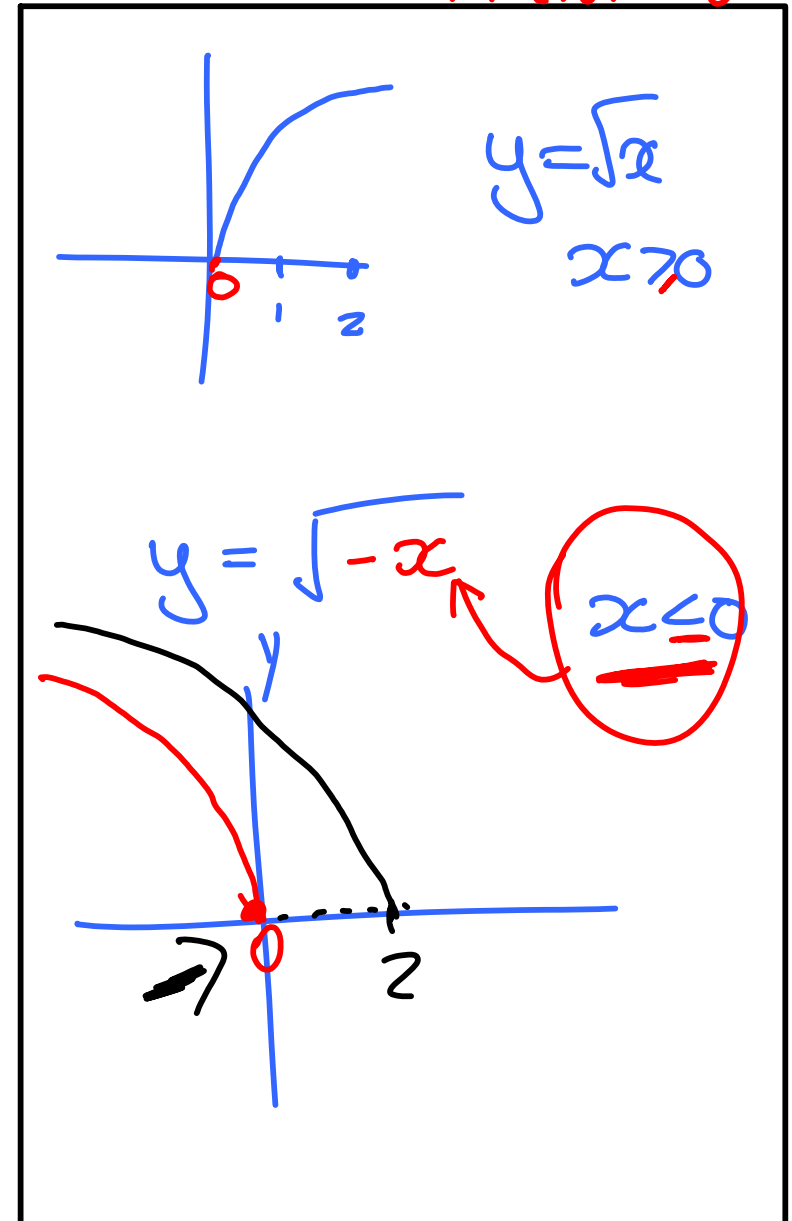
• Gebied = $D_f = \{x \mid x \leq 2\}$
 $= (-\infty, 2]$

• Skets: $y = \sqrt{2-x}$ $x \geq 0$

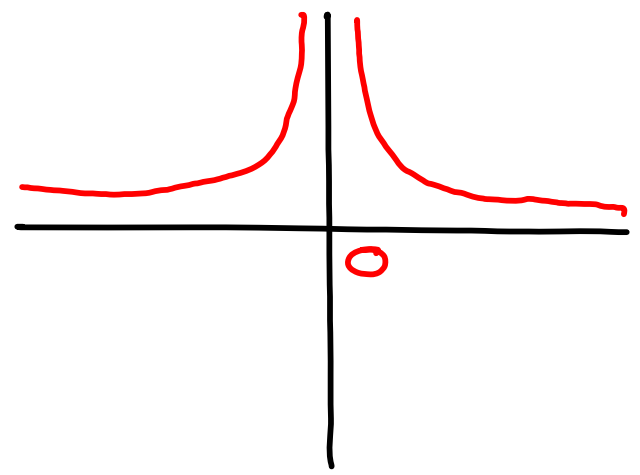
• $W_f = [0, \infty)$

• $f : D_f \rightarrow W_f$
 $f : (-\infty, 2] \rightarrow [0, \infty)$

Eenheid 1.5



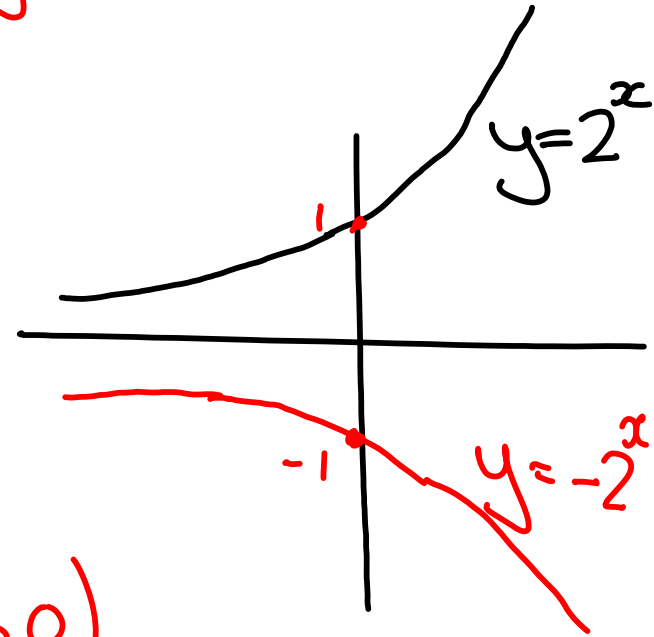
$$\textcircled{2} \quad g(t) = \frac{1}{t^2}$$



- gebied : $D_f = \{x \mid x \neq 0\}$
 $= (-\infty, 0) \cup (0, \infty)$
- Waardeverzameling = $(0, \infty)$
 $W_f = \{y \mid y > 0\}$
- $g: (-\infty, 0) \cup (0, \infty) \rightarrow (0, \infty)$

$$\textcircled{3} \quad y = -2^x = f(x)$$

- gebied = $\mathbb{R} = \{x \mid x \in \mathbb{R}\}$
- Waardeverzameling = $(-\infty, 0)$
- $f: \mathbb{R} \rightarrow (-\infty, 0)$



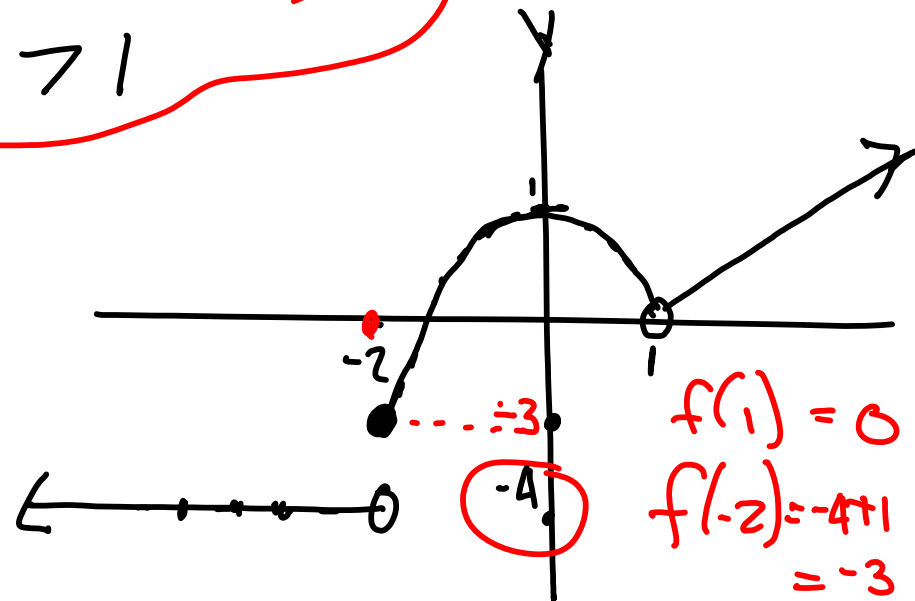
Stuksgewijs gedefinieerde functies

- gebied
- Skets f
- Waardeverzameling

$$\textcircled{1} \quad f(x) = \begin{cases} -4 & \text{as } x < -2 \\ -x^2 + 1 & \text{as } -2 \leq x < 1 \\ x - 1 & \text{as } x \geq 1 \end{cases}$$

Bepelings

$$\begin{aligned} \bullet \quad D_f &= \mathbb{R} \setminus \{1\} \\ &= (-\infty, 1) \cup (1, \infty) \end{aligned}$$



- $\omega_f = [-3, \infty) \cup \{-4\}$

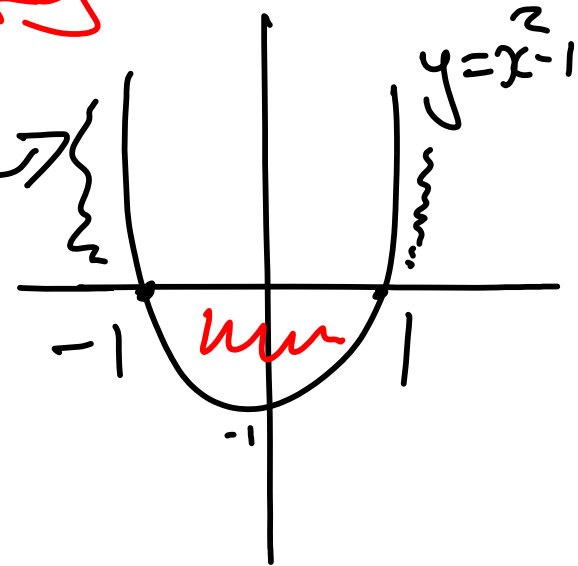
- $f : \overset{D_f}{(-\infty, 1) \cup (1, \infty)} \xrightarrow{\omega_f} \{-4\} \cup [-3, \infty)$

② $f(x) = |x^2 - 1| + 1$

Definisie = $\begin{cases} (x^2 - 1) + 1 & \text{as } \underline{x^2 - 1} \geq 0 \\ -(x^2 - 1) + 1 & \text{as } \underline{x^2 - 1} < 0 \end{cases}$

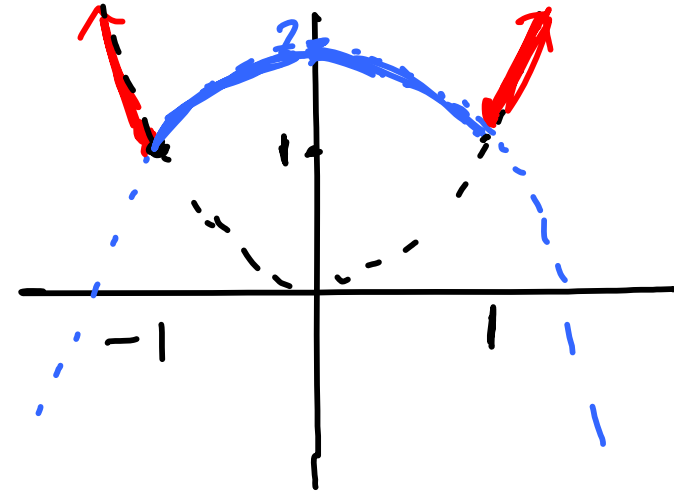
= $\begin{cases} x^2 & \text{as } \underline{x \geq 1 \text{ of } x \leq -1} \\ -x^2 + 2 & \text{as } \underline{-1 < x < 1} \end{cases}$

Beperting



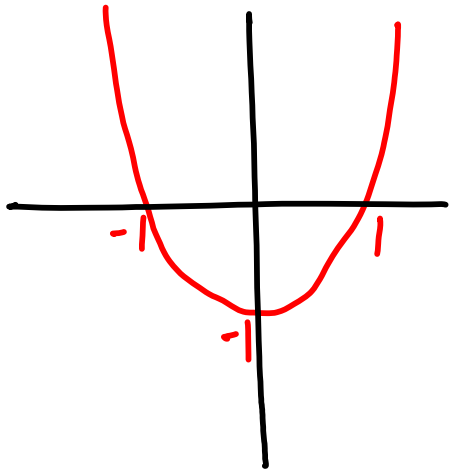
- $D_f = \mathbb{R}$

$$\bullet \omega_f = \{y \mid y \geq 1\} = [1, \infty)$$

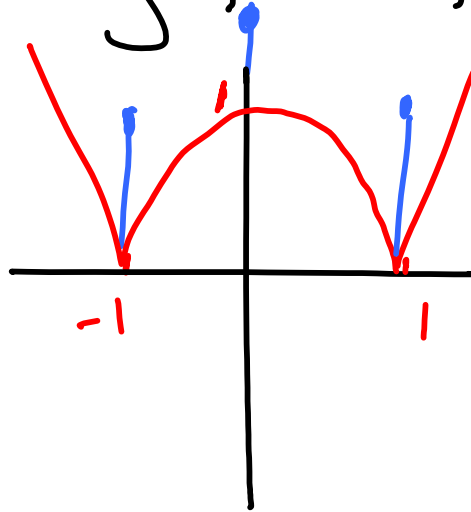


OF Skets $y = |x^2 - 1| + 1$

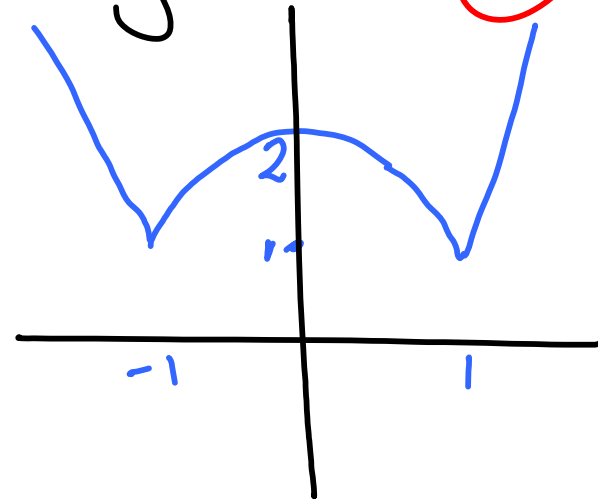
$$\bullet y = x^2 - 1$$



$$\bullet y = |x^2 - 1|$$



$$\bullet y = |x^2 - 1| + 1$$



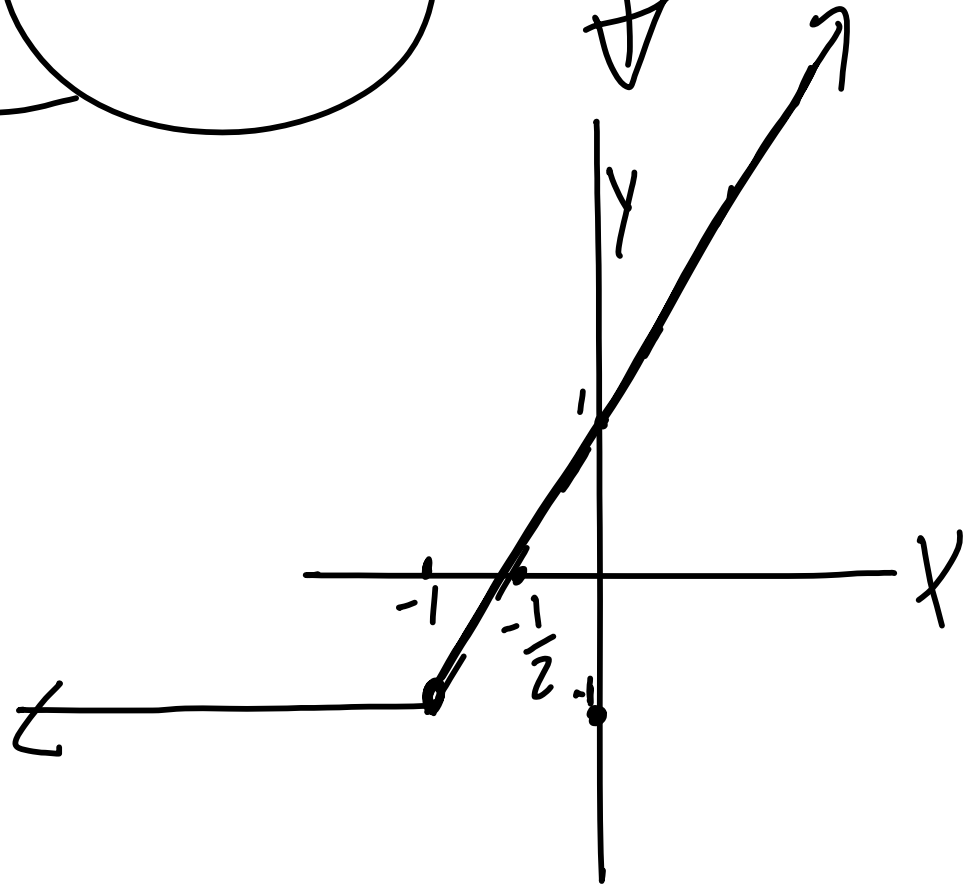
$$\textcircled{3} \quad f(x) = x + |x+1|$$

$$= \begin{cases} x + (x+1) & \text{as } x+1 \geq 0 \\ x - (x+1) & \text{as } x+1 < 0 \end{cases} \quad \text{Bepending}$$

$$= \begin{cases} 2x+1 & \text{as } x \geq -1 \\ -1 & \text{as } x < -1 \end{cases} \quad \text{Bepend}$$

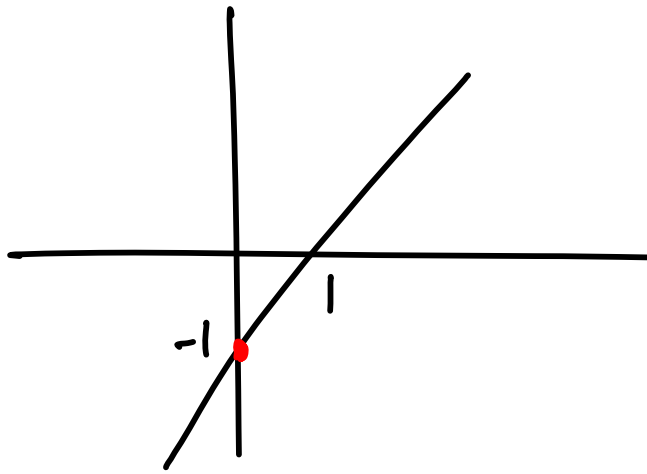
$$\bullet D_f = \mathbb{R}$$

$$\bullet W_f = [-1, \infty)$$

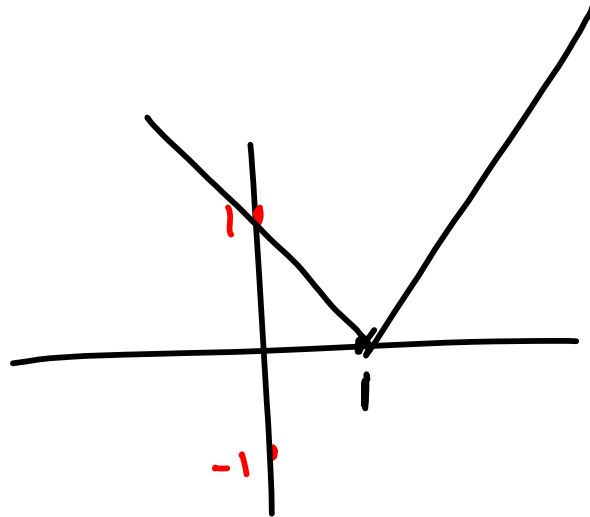


Skets grafieke $y = |f(x)|$ as jy
weet hoe lyk die grafiek van
 $y = f(x)$.

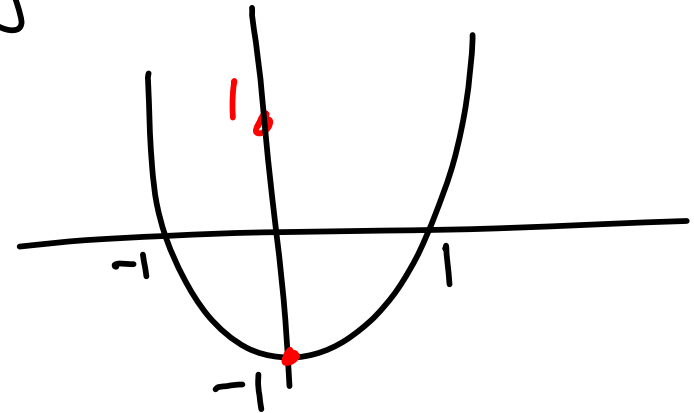
① $y = x - 1$



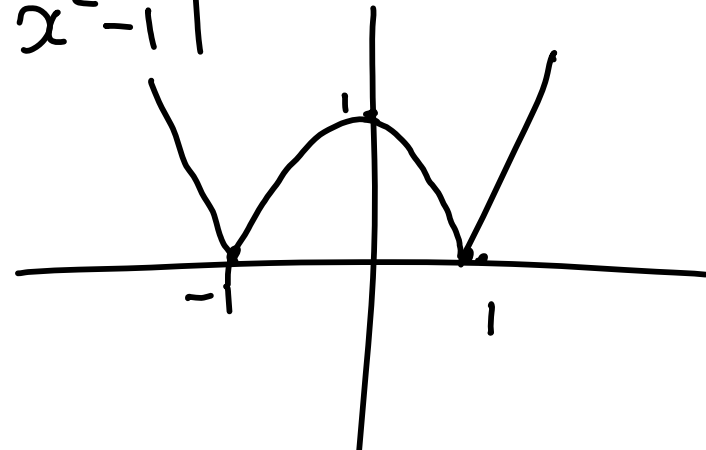
• $y = |x - 1|$



② $y = x^2 - 1$



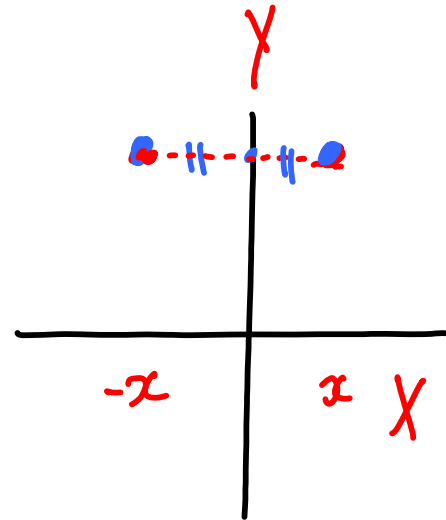
• $y = |x^2 - 1|$



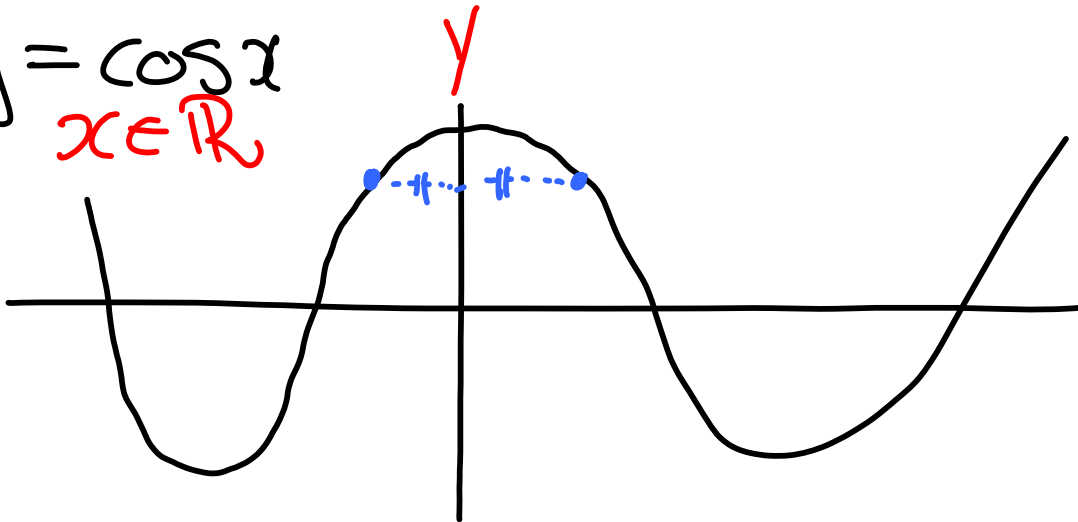
Simmetriese funksies Ewe / oneve funksies

Definisies: (ken volledig)

- ~~Def~~ f is 'n ewe funksie as $f(-x) = f(x)$ vir alle $x \in D_f$
Toetswaag 1 of 0



Bv. $y = \cos x$
 $x \in \mathbb{R}$

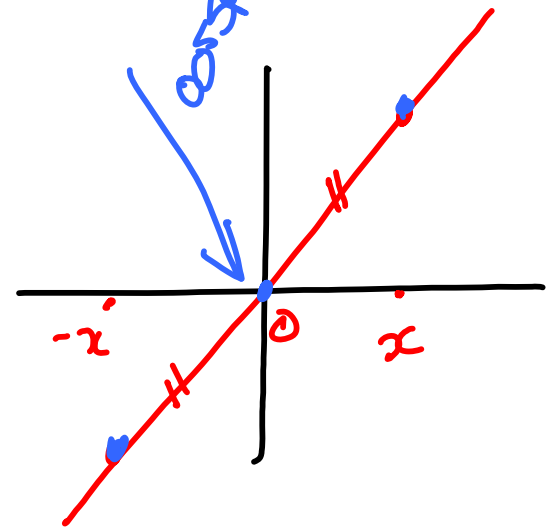


ewe funksie

• f is an odd function as

$$\underline{f(-x)} = -\underline{f(x)} \text{ for all } \underline{x \in D_f}$$

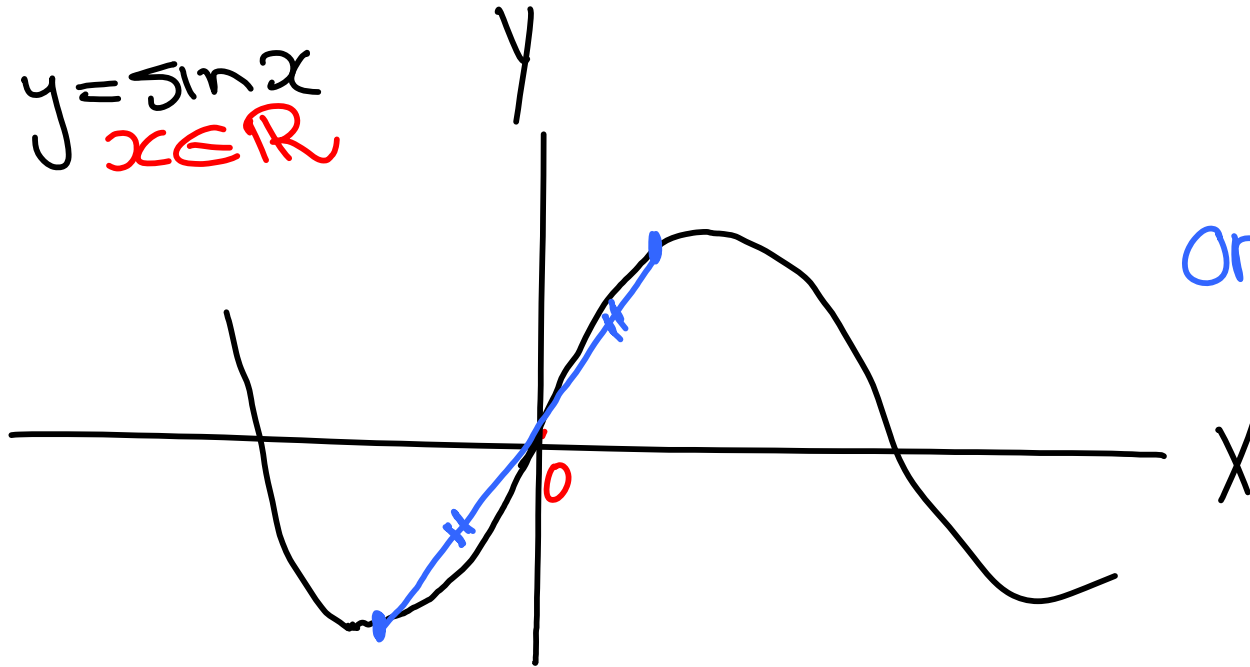
Verh.



Bv.

$$y = \sin x$$

$x \in \mathbb{R}$



odd function

Is die funksie ewe, oneve of
nie een van die twee nie?

① $g(x) = -x^2 + 4$ is ewe
 $g(\underline{-x}) = -(-x)^2 + 4$
 $= -x^2 + 4 = g(\underline{x})$

• $g(-x) = g(x)$
ewe

• $g(-x) = -g(x)$
oneve

• ?? Nie

② $f(t) = \frac{-t}{t^2 + 1}$ is oneve

$$f(-t) = \frac{-(-t)}{(-t)^2 + 1} = - \left[\frac{-t}{t^2 + 1} \right] = -f(t)$$

ewe
? $\stackrel{?}{=} f(t)$
? $\stackrel{?}{=} -f(t)$

$$\textcircled{3} \quad h(x) = x^2 + 2x + 1$$

$$h(-x) = (-x)^2 + 2(-x) + 1 = x^2 - 2x + 1$$

Nou: $h(x) \neq h(x)$ nie ewe

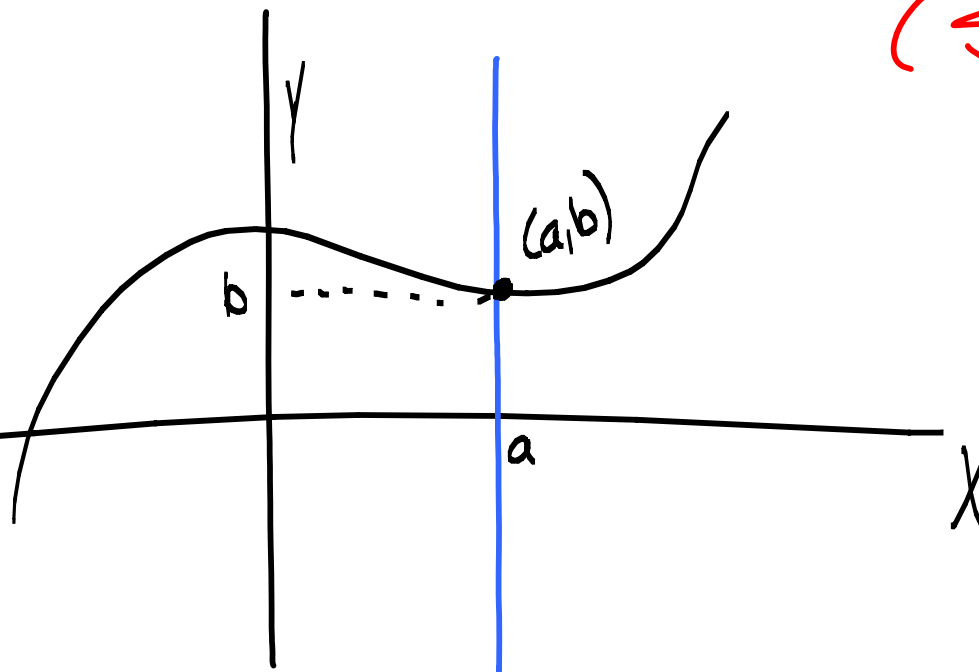
$h(-x) \neq -h(x)$ nie onewe nie

Vertikale lyntoets (funksie of nie)

(Stewart p15)

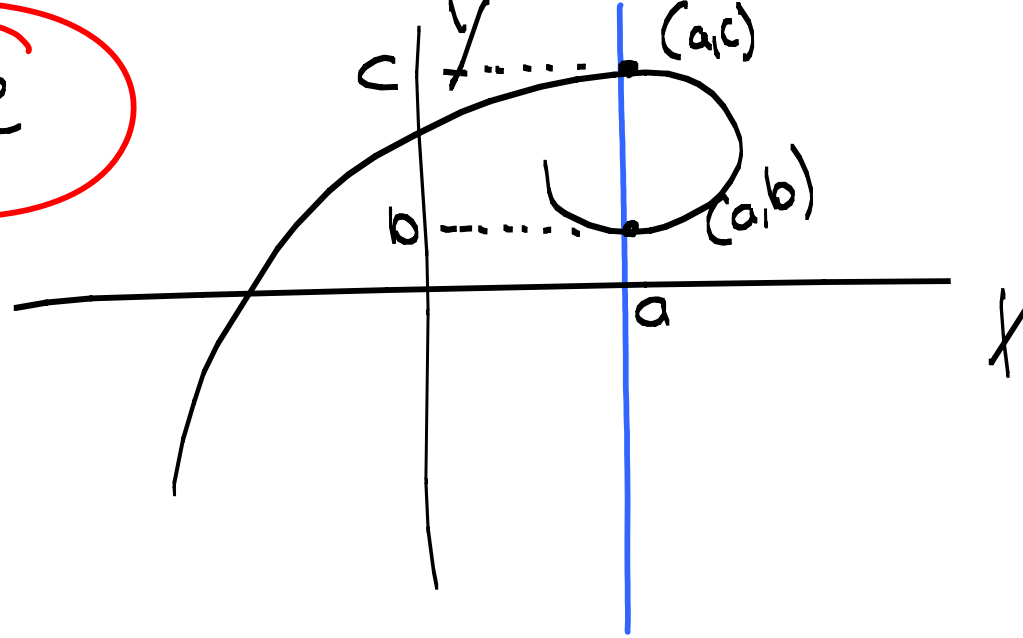
① Ja / Nee

funksie?



② Ja / Nee

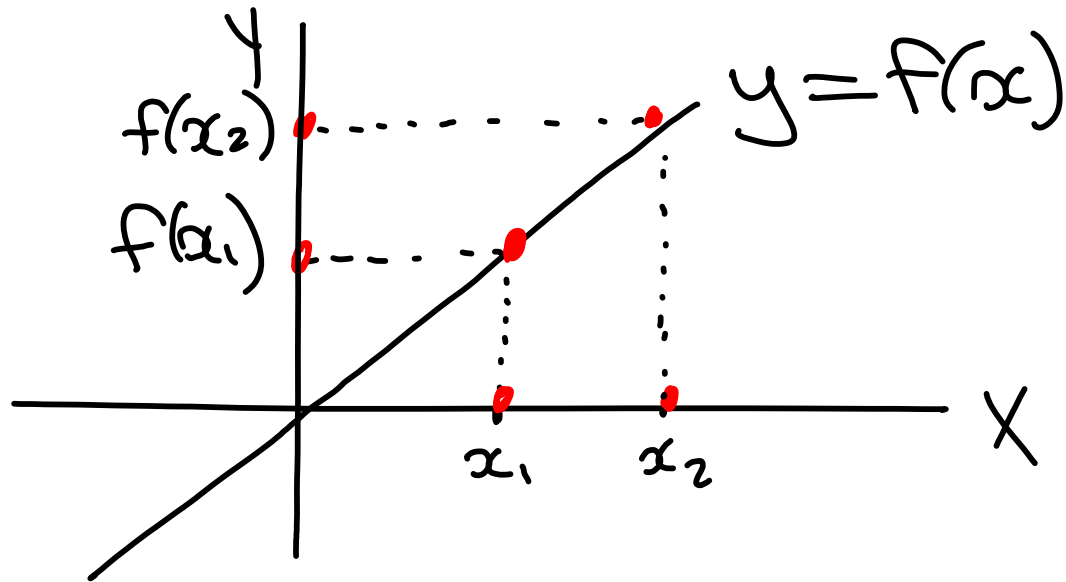
Funksie?



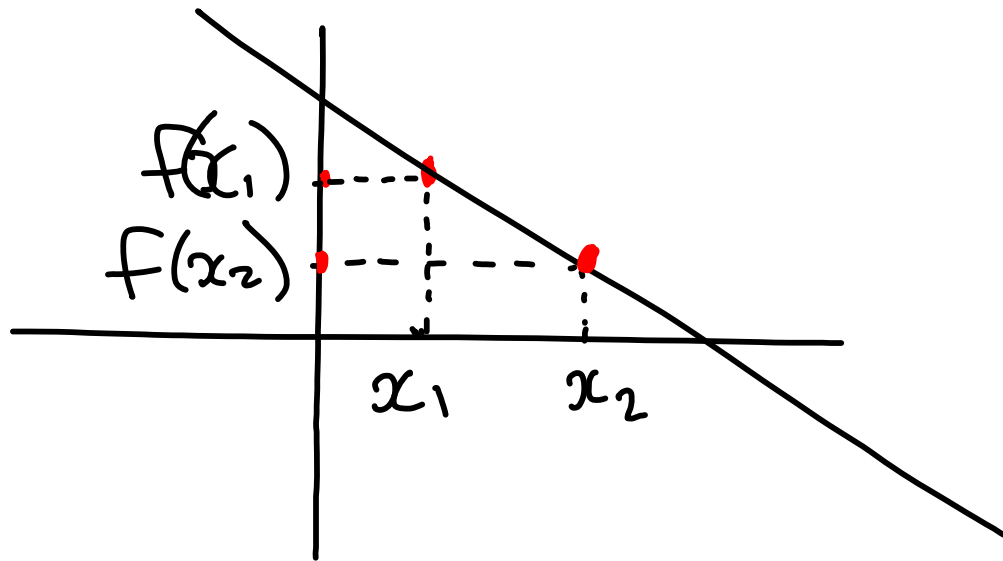
• Stygende / Dalende Funksies. B

Definisie (ken volledig)

f styg op interval I
as $f(x_1) < f(x_2)$ wanneer $x_1 < x_2$ in I



f daal op in interval I
 as $f(x_1) > f(x_2)$ wanneer $x_1 < x_2$ in I



VBD $f(x) = x^2$

f styg op $(0, \infty)$

f daal op $(-\infty, 0)$

